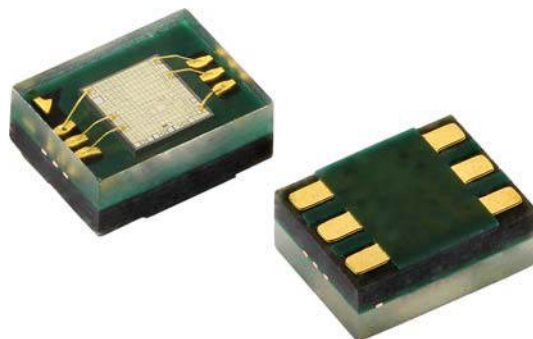


UV A Light Sensor with I²C Interface



DESCRIPTION

VEML6070 is an advanced ultraviolet (UV) light sensor with I²C protocol interface and designed by the CMOS process. It is easily operated via a simple I²C command. The active acknowledge (ACK) feature with threshold windows setting allows the UV sensor to send out a UVI alert message. Under a strong solar UVI condition, the smart ACK signal can be easily implemented by the software programming.

VEML6070 incorporates a photodiode, amplifiers, and analog / digital circuits into a single chip. VEML6070's adoption of Filtron™ UV technology provides the best spectral sensitivity to cover UV spectrum sensing. It has an excellent temperature compensation and a robust refresh rate setting that does not use an external RC low pass filter. VEML6070 has linear sensitivity to solar UV light and is easily adjusted by an external resistor. Software shutdown mode is provided, which reduces power consumption to be less than 1 µA. VEML6070's operating voltage ranges from 2.7 V to 5.5 V.

FEATURES

- Package type: surface mount
- Dimensions (L x W x H in mm): 2.35 x 1.8 x 1.0
- Integrated modules: ultraviolet sensor (UV), and signal conditioning IC
- Converts solar UV light intensity to digital data
- Excellent UV sensitivity and linearity via Filtron™ technology
- Excellent performance of UV radiation measurement under long time solar UV exposure
- Excellent temperature compensation
- High dynamic detection resolution
- Standard I²C protocol interface
- Support acknowledge feature (ACK)
- Immunity on fluorescent light flicker software shutdown mode control
- Package: OPLGA
- Temperature compensation: -40 °C to +85 °C
- Floor life: 168 h, MSL 3, according to J-STD-020
- Output type: I²C bus
- Operation voltage: 2.7 V to 5.5 V
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



APPLICATIONS

- Solar UV indicator
- Cosmetic / outdoor sport handheld product
- Consumer products

PRODUCT SUMMARY

PART NUMBER	OPERATING VOLTAGE RANGE (V)	I ² C BUS VOLTAGE RANGE (V)	PEAK SENSITIVITY (nm)	RANGE OF SPECTRAL BANDWIDTH λ _{0.5} (nm)	OUTPUT CODE
VEML6070	2.7 to 5.5	1.7 to 5.5	355	± 20	16 bit, I ² C

Note

(1) Adjustable through I²C interface

ORDERING INFORMATION

ORDERING CODE	PACKAGING	VOLUME ⁽¹⁾	REMARKS
VEML6070	Tape and reel	MOQ: 2500 pcs	2.35 mm x 1.8 mm x 1.0 mm

Note

(1) MOQ: minimum order quantity

ABSOLUTE MAXIMUM RATINGS (T_{amb} = 25 °C, unless otherwise specified)

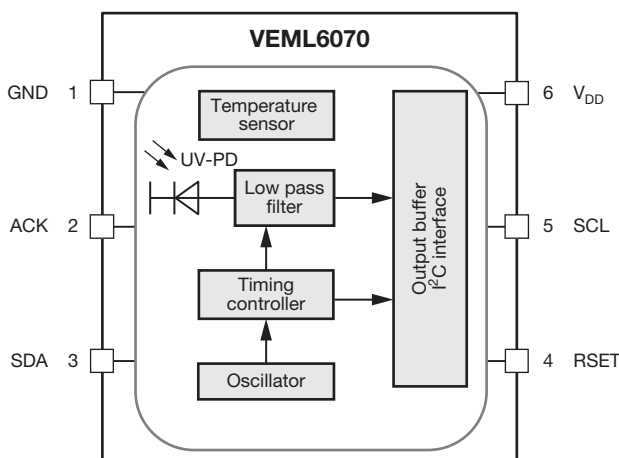
PARAMETER	TEST CONDITION	SYMBOL	MIN.	MAX.	UNIT
Supply voltage		V _{DD}	0	6.0	V
Operation temperature range		T _{amb}	-40	+85	°C

RECOMMENDED OPERATING CONDITIONS ($T_{amb} = 25^{\circ}\text{C}$, unless otherwise specified)

PARAMETER	TEST CONDITION	SYMBOL	MIN.	MAX.	UNIT
Supply voltage		V_{DD}	2.7	5.5	V
Operation temperature range		T_{amb}	-40	+85	$^{\circ}\text{C}$
I ² C bus operating frequency		$f_{(I2CCLK)}$	10	400	kHz

PIN DESCRIPTIONS

PIN ASSIGNMENT	SYMBOL	TYPE	FUNCTION
1	GND	I	Power supply ground, all voltage are reference to GND
2	ACK	O (open drain)	Acknowledge pin
3	SDA	I / O (open drain)	I ² C digital serial data output to the host
4	SET		Light reading adjustment, connect a resistor to GND
5	SCL	I	I ² C digital serial clock input from the host
6	V_{DD}	I	Supply voltage

BLOCK DIAGRAM

BASIC CHARACTERISTICS ($T_{amb} = 25^{\circ}\text{C}$, unless otherwise specified)

PARAMETER	TEST CONDITION	SYMBOL	MIN.	TYP.	MAX.	UNIT
Supply operation voltage		V_{DD}	2.7	-	5.5	V
Supply current	$R_{SET} = 240\text{ k}\Omega$ (1)(2)	I_{DD}	-	100	250	μA
I ² C signal input	Logic high	V_{IH}	1.5	-	V_{DD}	V
	Logic low	V_{IL}	-	-	0.8	
Peak sensitivity wavelength		λ_p	-	355	-	nm
Range of spectral sensitivity		$\lambda_{0.1}$	320	-	410	nm
UVA sensitivity	$R_{SET} = 240\text{ k}\Omega$, $I_T = 1\text{ T}$ (3)		-	5	-	$\mu\text{W}/\text{cm}^2/\text{step}$
Maximum UVA detection power	$R_{SET} = 240\text{ k}\Omega$, $I_T = 1\text{ T}$ (3)		-	-	328	mW/cm^2
Dark offset	$R_{SET} = 240\text{ k}\Omega$, $I_T = 1\text{ T}$ (1)		0	1	5	steps
Output offset	$R_{SET} = 240\text{ k}\Omega$, $I_T = 1\text{ T}$ (1)(4)		-	2	-	steps
Shutdown current	Light condition = dark (1)	I_{DD}	-	1	15	μA

Notes

- (1) Test condition: $V_{DD} = 3.3\text{ V}$, temperature: 25°C
- (2) Light source: solar light source
- (3) Test using 365 nm UVA LED
- (4) Ambient light intensity = 500 lx

I²C TIMING CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)						
PARAMETER	SYMBOL	STANDARD MODE		FAST MODE		UNIT
		MIN.	MAX.	MIN.	MAX.	
Clock frequency	$f_{(SMBCLK)}$	10	100	10	400	kHz
Bus free time between start and stop condition	$t_{(BUF)}$	4.7	-	1.3	-	μs
Hold time after (repeated) start condition; after this period, the first clock is generated	$t_{(HDSTA)}$	4.0	-	0.6	-	μs
Repeated start condition setup time	$t_{(SUSTA)}$	4.7	-	0.6	-	μs
Stop condition setup time	$t_{(SUSTO)}$	4.0	-	0.6	-	μs
Data hold time	$t_{(HDDAT)}$		3450	-	900	ns
Data setup time	$t_{(SUDAT)}$	250	-	100	-	ns
I ² C clock (SCK) low period	$t_{(LOW)}$	4.7	-	1.3	-	μs
I ² C clock (SCK) high period	$t_{(HIGH)}$	4.0	-	0.6	-	μs
Detect clock / data low timeout	$t_{(TIMEOUT)}$	25	35	-	-	ms
Clock / data fall time	$t_{(F)}$	-	300	-	300	ns
Clock / data rise time	$t_{(R)}$	-	1000	-	300	ns

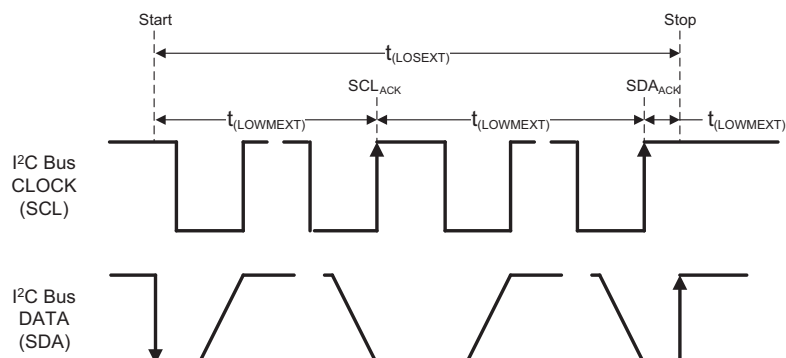
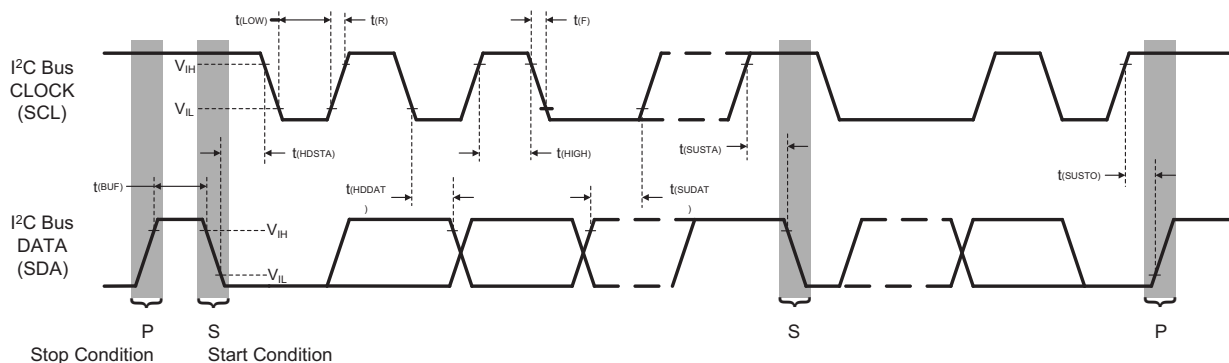


Fig. 1 - I²C Timing Diagram

PARAMETER TIMING INFORMATION

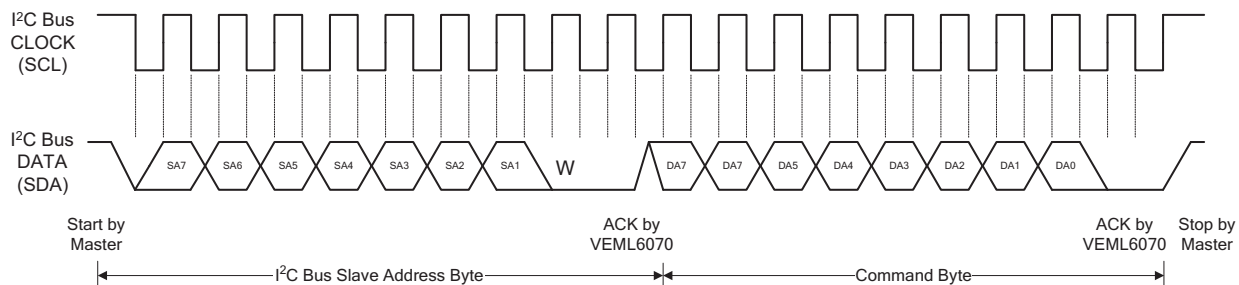


Fig. 2 - Timing for Send Byte Command Format

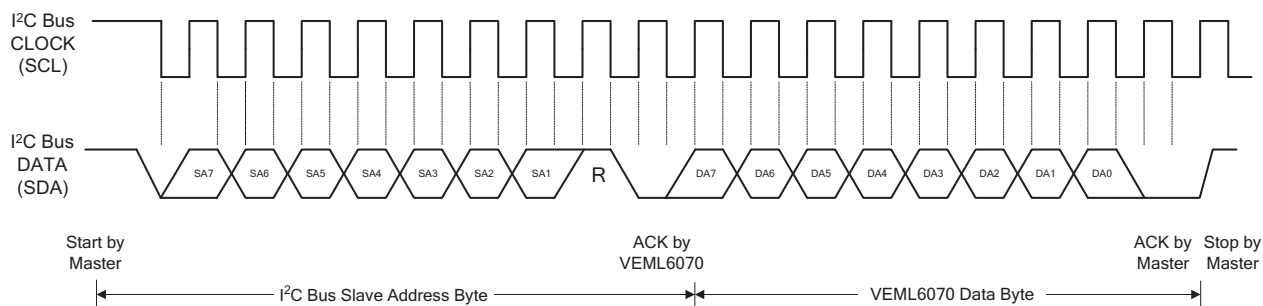


Fig. 3 - I²C Timing for Receive Byte Command Format

TYPICAL PERFORMANCE CHARACTERISTICS ($T_{amb} = 25^{\circ}\text{C}$, unless otherwise specified)

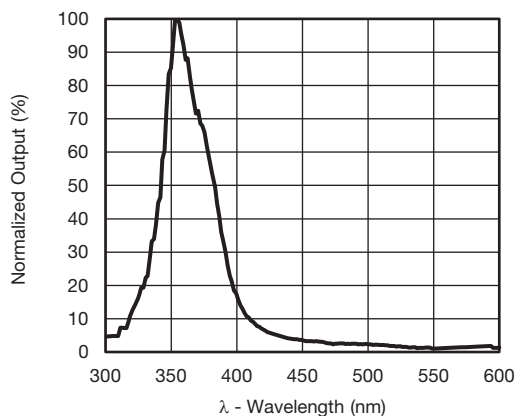


Fig. 4 - Normalized Spectral Response

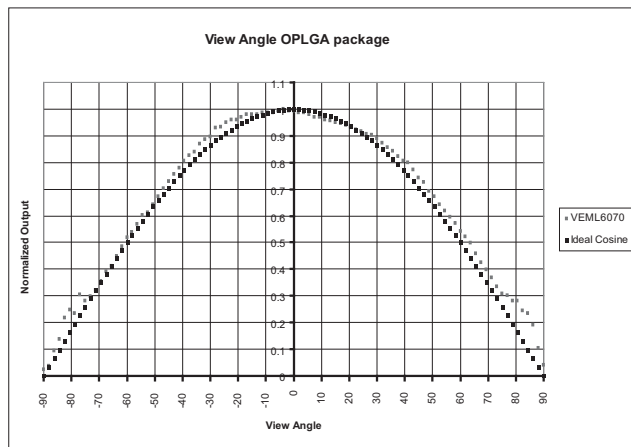


Fig. 5 - Normalized Output vs. View Angle

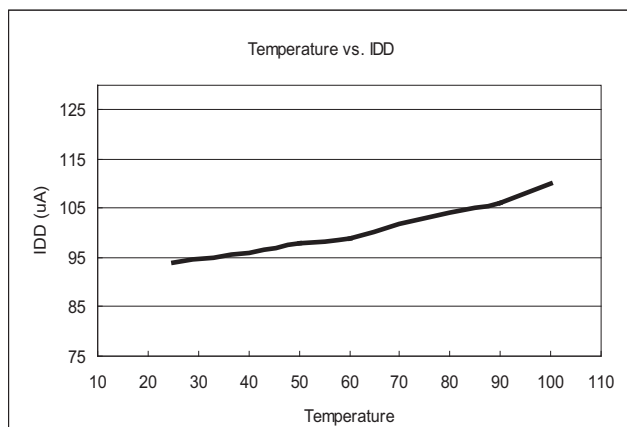
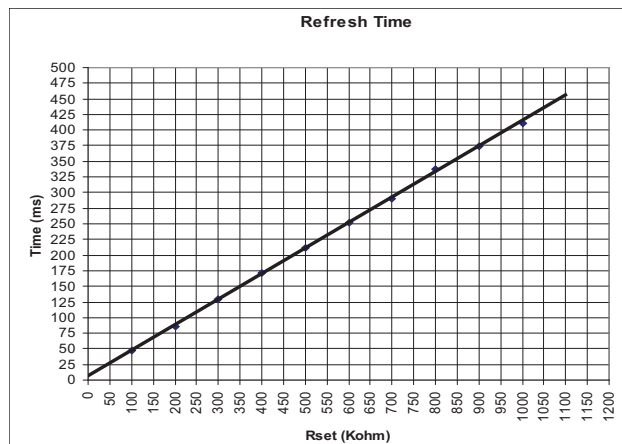

Fig. 6 - I_{DD} vs. Temperature


Fig. 7 - Refresh Time

APPLICATION INFORMATION

Pin Connection with the Host

VEML6070 is a cost effective solution for ultraviolet light sensing with I²C interface. The standard serial digital interface easily accesses “UV light intensity” digital data.

The additional capacitor near the V_{DD} pin is used for power supply noise rejection. For the I²C bus design, the pull-up voltage refers to the I/O of the baseband due to the “open drain” design. The pull-up resistors for the I²C bus design are recommended to be 2.2 kΩ. The circuit diagram as an example is shown in figure 8.

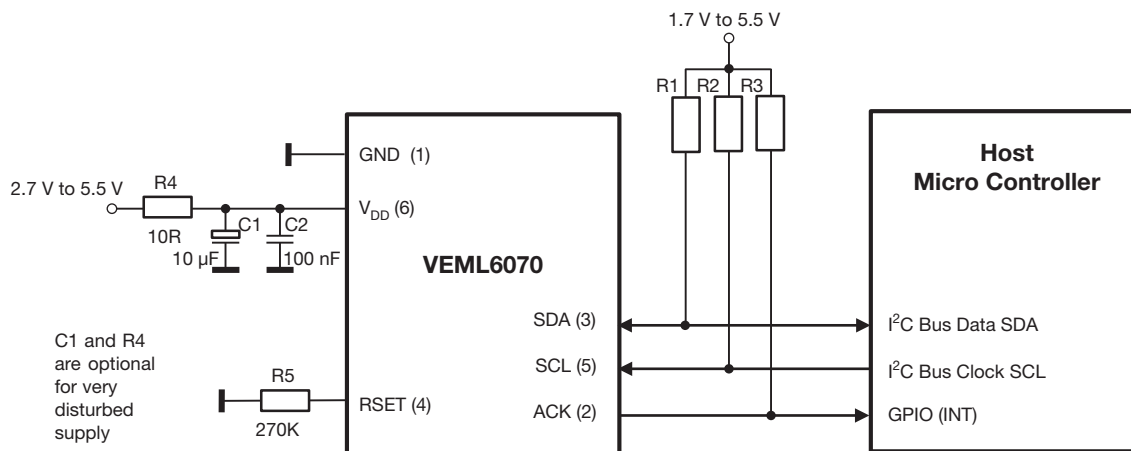


Fig. 8 - Hardware Pin Connection Diagram

Digital Interface

VEML6070 contains a 8-bit command register written via the I²C bus. All operations can be controlled by the command register. The simple command structure enables users to easily program the operation setting and latch the light data from VEML6070. In figure 9, VEML6070 I²C command format description for reading and writing operation between the host and VEML6070 are shown. The white sections indicate host activity and the gray sections indicate VEML6070's acknowledgement of the host access activity.

Receive byte → read data from UVS



Send byte → write command to UVS



S = start condition

P = stop condition

A = acknowledge

Shaded area = VEML6070 acknowledge

Fig. 9 - VEML6070 Command Protocol

Slave Address and Function Description

The VEML6070 has one slave address used for write functions (command) and two slave addresses used for read functions (UV data LSB and MSB).

The 7-bit address for write functions is 38h = 0111000x resulting in a 70h = 01110000 8-bit address. The 7-bit addresses for read functions are 38h = 0111000x for the UV Data LSB and 39h = 0111001x for the UV data MSB. This results in a 71h = 01110001 and 73h = 01110011 8-bit address, respectively. The 7-bit address 39h should not be used for a write function.

Command Register Format

VEML6070 provides a command to set device operations and sensitivity adjustment. This command is 8-bit long and includes 4 parameter groups for programming. The command format descriptions and register setting explanations are shown in tables 1 and 2.

TABLE 1 - COMMAND REGISTER BITS DESCRIPTION							
COMMAND FORMAT							
Reserved		ACK	ACK_THD	IT		Reserved	SD
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
0	0	ACK	THD	IT1	IT0	1	SD
DESCRIPTION							
Reserved		Reserved					
ACK		Acknowledge activity setting					
ACK_THD		Acknowledge threshold window setting for byte mode usage					
IT		Integration time setting					
SD		Shutdown mode setting					

TABLE 2 - REGISTER TABLE SETTING

BITS SETTING		DESCRIPTION		
Reserved	Set initial value to (0 : 0)		(IT1 : IT0) ⁽¹⁾ (0 : 0) = ½T (0 : 1) = 1T (1 : 0) = 2T (1 : 1) = 4T	
ACK	0 = disable		Reserved	Set initial value to 1
	1 = enable			
ACK_THD	0 = 102 steps		SD	0 = disable
	1 = 145 steps			1 = enable

Note

(1) Please refer to table 4, “Example of Refresh Time and R_{SET} Value Relation”

Data Access

VEML6070 has 16-bit resolution to give high resolution for light intensity sensing. Examples of the application setting are shown in table 3.

TABLE 3 - DATA ACCESS DESCRIPTION

	VEML6070 16-BIT DATA BUFFER															
Data bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Sequence 1																
Sequence 2																

Notes

- Slave addresses (8 bits) for data read: 0x71 and 0x73
- Data reading sequence for the host:
 - Set read command to 0x73, read MSB 8 bits of 16 bits light data (sequence 1)
 - Set read command to 0x71, read LSB 8 bits of 16 bits light data for completing data structure (sequence 2)

Initialization

VEML6070 needs to be initialized while the system's power is on. The initialization includes two major steps: (1) clear ACK state of UVS and (2) fill the initial value, 06 (HEX), into the 0x70 addresses. After the initialization is completed, VEML6070 can be programmable for operation by write command setting from the host. VEML6070 initialization is recommended to be completed within 150 ms.

Acknowledge Activity

VEML6070 provides a function for sending an acknowledge signal (ACK) to the host when the value of sensed UV light is over the programmed threshold (ACK_THD) value. The purpose of the ACK signal is similar to the interrupt feature which informs the host once the sensed data level goes beyond the interrupt threshold setting. VEML6070 has two ACK threshold values, 102 steps and 145 steps.

There are two methods of driving acknowledge condition and read / write command to VEML6070:

- (1) If the host implements the INT function, it performs a modified received byte operation to disengage VEML6070's acknowledge signal and acknowledge alert response address (ARA), 0x18 (Hex). A command format for responses to an ARA is shown in figure 10.



Fig. 10 - Command Format for Responds to an ARA

- (2) If the host does not implement this feature, it should periodically access the ARA or read ARA before setting each read / write command.

The behavior of an ACK signal is similar to the INT definition in I²C specification. For the hardware circuit design, this pin connects to an INT pin or GPIO pin of the MCU. The threshold ACK_THD definition is based on the sensitivity setting of VEM16070.



The ACK or UVI interrupt function allows the UVI sensing system to perform data pooling based on the interrupt event. The system sensor manager does not need to do continual data pooling and this significantly reduced the MCU loading. The ACK signal can also be used as a trigger event for popping up a warning UVI message.

Refresh Time Determination

VEML6070's refresh time can be determined by the R_{SET} value. Cooperating with the command register setting, the designer has a flexible way of defining the timing for light data collection. The default refresh time is 1T, (IT1 : IT0) = (0 : 1). If the R_{SET} value is changed, the default timing changes and the other parts in the register table also change by comparing itself with the default timing (refer to figure 7).

Table 4 is an example of two R_{SET} resistors that show the timing table that the system designer can use a flexible way to determine the desired refresh time.

TABLE 4 - EXAMPLE OF REFRESH TIME AND R_{SET} VALUE RELATION			
REGISTER	SETTING	REFRESH TIME	
		$R_{SET} = 300\text{ k}\Omega$	$R_{SET} = 600\text{ k}\Omega$
(IT1 : IT0)	(0 : 0) = $\frac{1}{2}T$	62.5 ms	125 ms
	(0 : 1) = 1T	125 ms	250 ms
	(1 : 0) = 2T	250 ms	500 ms
	(1 : 1) = 4T	500 ms	1000 ms

The designer can decide the refresh timing range requirement first, then choose an appropriate R_{SET} value for the timing range, and then write the correct value for the system application via I²C protocol.

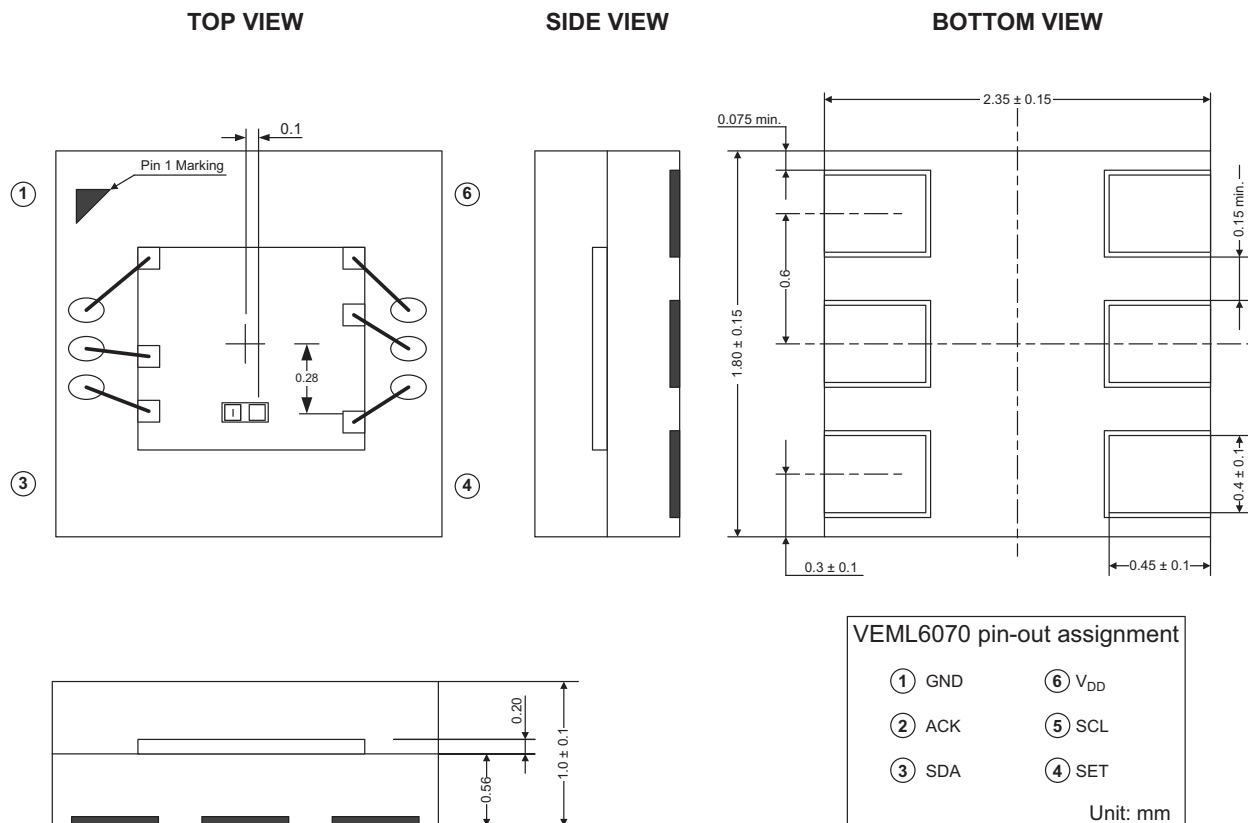
PACKAGE INFORMATION in millimeters


Fig. 11 - VEML6070 A3OP Package Dimensions

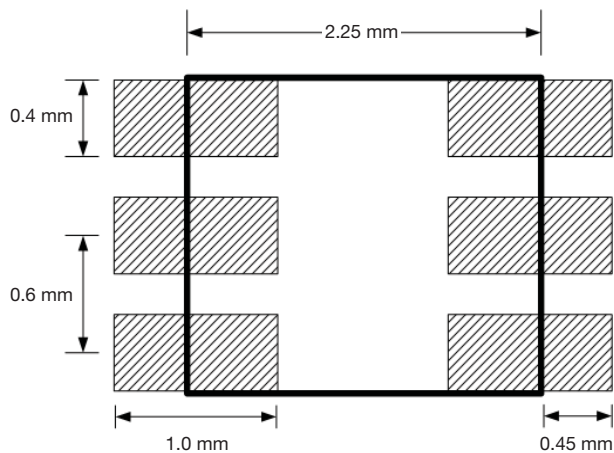
LAYOUT NOTICE


Fig. 12 - VEML6070 OPLGA PCB Layout Footprint

APPLICATION CIRCUIT BLOCK REFERENCE

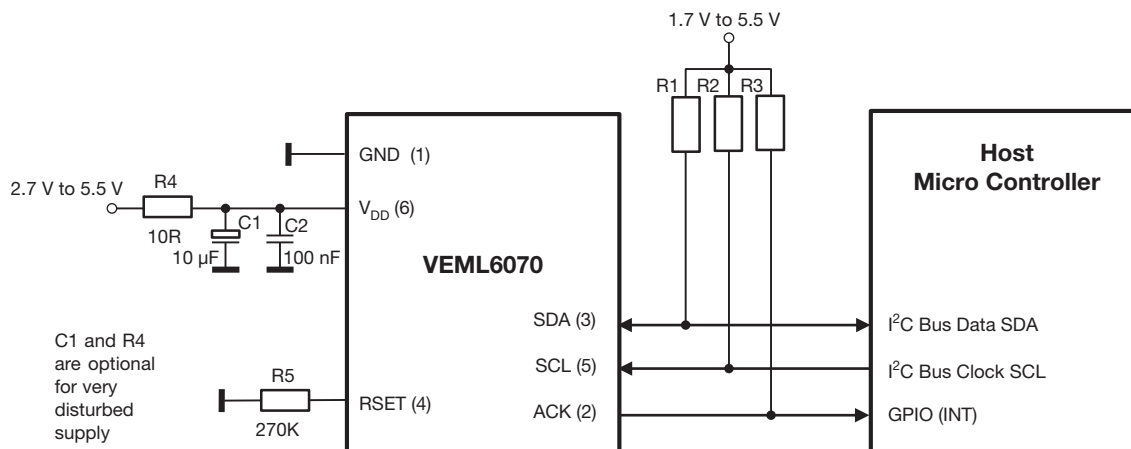


Fig. 13 - VEML6070 Application Circuit

Notes

- V_{DD} range: 2.7 V to 5.5 V
- The pull-up voltage for I²C bus is referring to the I/O specification of baseband

RECOMMENDED STORAGE AND REBAKING CONDITIONS				
PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
Storage temperature		5	50	°C
Relative humidity		-	60	%
Open time	Rebaking process should be done when aluminized envelope reopened	-	-	
Total time	From the date code on the aluminized envelope (unopened)	-	6	months
Rebaking	Tape and reel: 60 °C	-	22	h
	Tube: 60 °C	-	22	h

RECOMMENDED INFRARED REFLOW

Soldering conditions are based on J-STD-020 C definition.

1. After opening the tape and reel, IR reflow process should be done
2. IR reflow profile conditions

IR REFLOW PROFILE CONDITION			
PARAMETER	CONDITIONS	TEMPERATURE	TIME
Peak temperature		255 °C + 0 °C / - 5 °C (max.: 260 °C)	10 s
Preheat temperature range and timing		150 °C to 200 °C	60 s to 180 s
Timing within 5 °C to peak temperature		-	10 s to 30 s
Timing maintained above temperature / time		217 °C	60 s to 150 s
Timing from 25 °C to peak temperature		-	8 min (max.)
Ramp-up rate		3 °C/s (max.)	-
Ramp-down rate		6 °C/s (max.)	-

3. Recommend Normal Solder Reflow is 235 °C to 255 °C

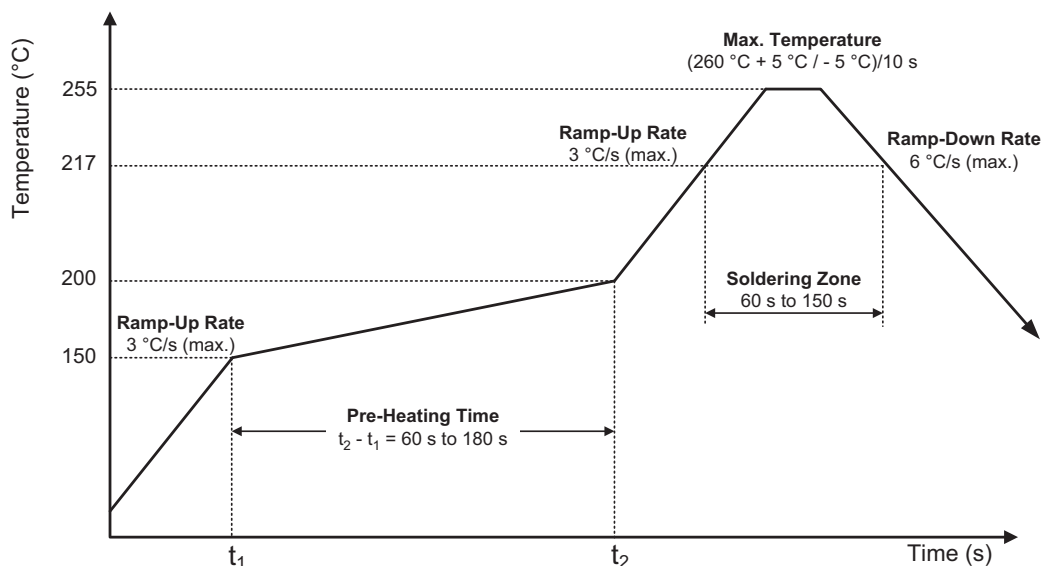


Fig. 14 - VEML6070 A3OP Solder Reflow Profile Chart

RECOMMENDED IRON TIP SOLDERING CONDITION AND WARNING HANDLING

1. Solder the device with the following conditions:
 - 1.1. Soldering temperature: 400 °C (max.)
 - 1.2. Soldering time: 3 s (max.)
2. If the temperature of the method portion rises in addition to the residual stress between the leads, the possibility that an open or short circuit occurs due to the deformation or destruction of the resin increases.
3. The following methods: VPS and wave soldering, have not been suggested for the component assembly.
4. Cleaning method conditions:
 - 4.1. Solvent: methyl alcohol, ethyl alcohol, isopropyl alcohol
 - 4.2. Solvent temperature < 45 °C (max.)
 - 4.3. Time: 3 min (min.)

TAPE PACKAGING INFORMATION in millimeters

DIMENSION OF CARRIER TAPE

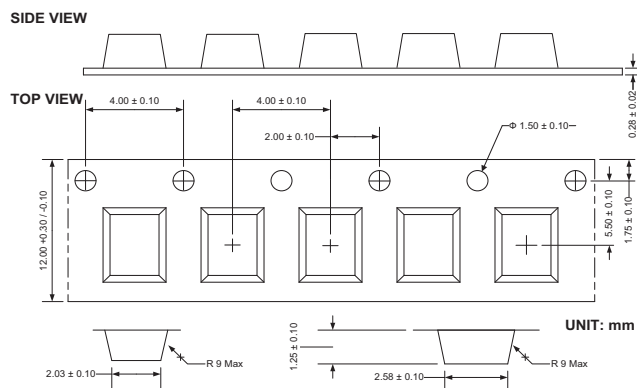


Fig. 15 - VEML6070 A3OP Package Carrier Tape

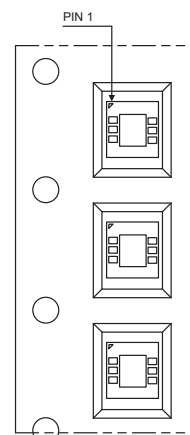


Fig. 16 - Taping Direction

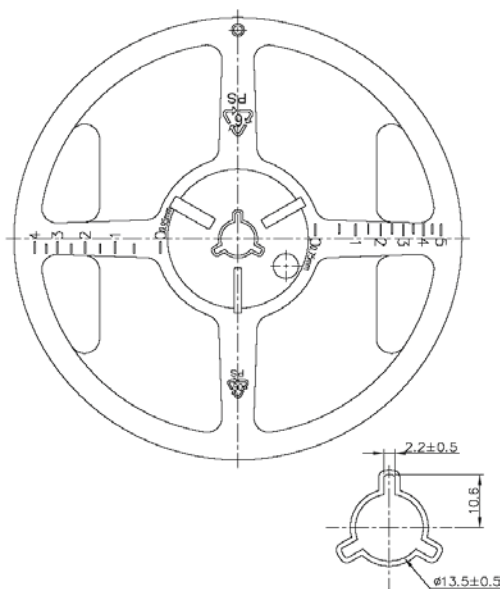
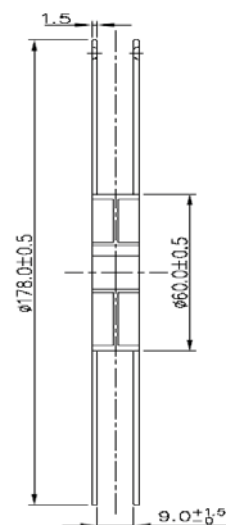


Fig. 17 - Reel Dimension





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